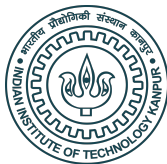


Lecture 6

Multivariate Posterior Summarization

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(Edited from the lecture notes by Brian J. Reich, NC State)



Summarizing multivariate posteriors

- ▶ A univariate posterior can be fully understood by a simple plot.
- ▶ However, when there are many parameters, this is impossible.
- ▶ Say, $\theta = (\theta_1, \dots, \theta_p)$.
- ▶ Ideally, we reduce to the univariate marginal posteriors

$$p(\theta_1|Y) = \int \dots \int p(\theta_1, \dots, \theta_p|Y) d\theta_2, \dots, d\theta_p.$$

- ▶ The same ideas we used for univariate models are now applicable.
- ▶ However, computing these integrals is often challenging.

Bayesian one-sample t -test

- ▶ In this section, we will study the one-sample t -test in depth.
- ▶ Likelihood: $Y_i | \mu, \sigma \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$, $i = 1, \dots, n$.
- ▶ Priors: $\pi(\mu, \sigma^2) = \pi_1(\mu)\pi_2(\sigma^2)$, $\mu \sim \mathcal{N}(\mu_0, \sigma_0^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.
- ▶ The joint (bivariate PDF) of (μ, σ^2) is proportional to
$$\left\{ \sigma^{-n} \exp \left[-\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2\sigma^2} \right] \right\} \times \exp \left[-\frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right] \times (\sigma^2)^{-a-1} \exp \left[-\frac{b}{\sigma^2} \right].$$
- ▶ How to summarize this complicated function?

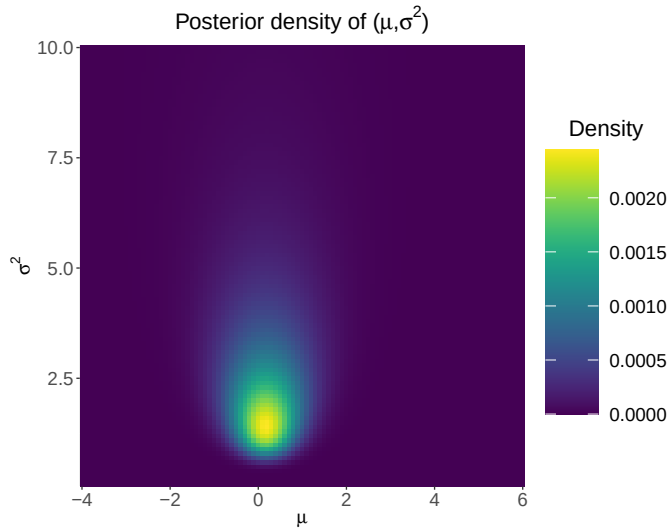
Plotting the posterior on a grid

- ▶ For models with only a few parameters, we could simply plot the posterior on a grid.
- ▶ That is, suppose, we compute $p(\mu, \sigma^2 | Y_1, \dots, Y_n)$ for all combinations of m values of μ and m values of σ^2 .
- ▶ The number of grid points is m^p where p is the number of parameters in the model.
- ▶ The posterior is plotted on the next slide for

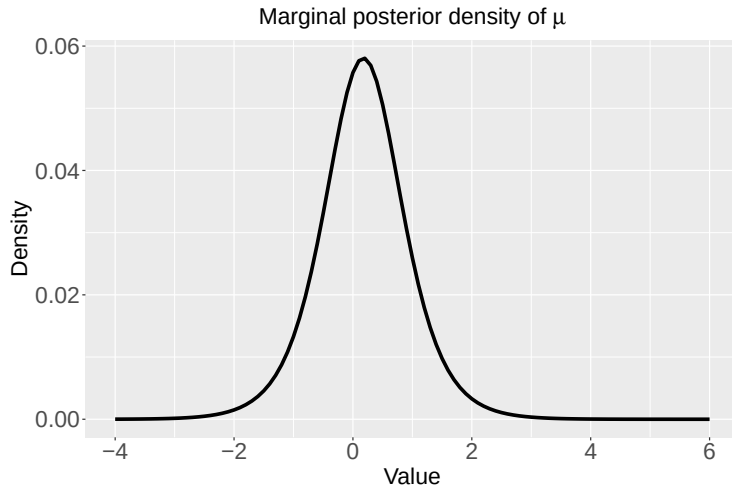
$$Y_1 = 2.68, Y_2 = 1.18, Y_3 = -0.97, Y_4 = -0.98, Y_5 = -1.03,$$

and priors $\mu | \sigma^2 \sim N(0, 100\sigma^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$.

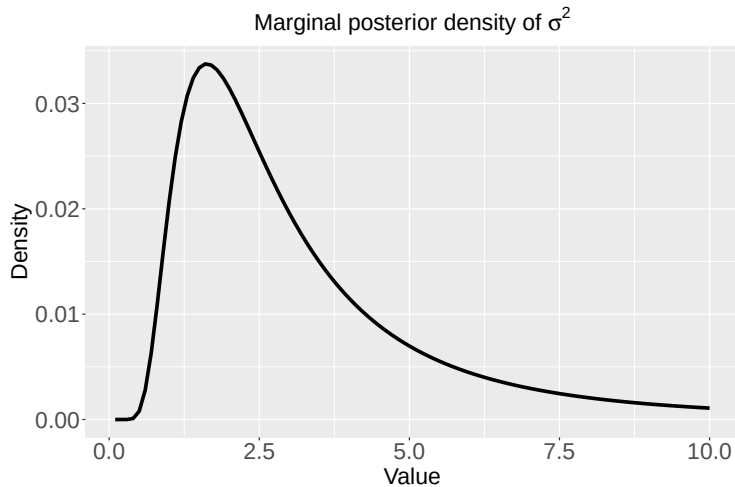
Bivariate posterior



Marginal posterior



Marginal posterior



Summarizing the results in a table

- ▶ Typically, we are interested in the marginal posterior

$$f(\mu|\mathbf{Y}) = \int_0^\infty p(\mu, \sigma^2|\mathbf{Y})d\sigma^2,$$

where $\mathbf{Y} = (Y_1, \dots, Y_n)$.

- ▶ This accounts for our uncertainty about σ^2 .
- ▶ We could also report the marginal posterior of σ^2 .
- ▶ Results are usually given in a table with marginal mean, SD, and 95% interval for all parameters of interest.
- ▶ The marginal posteriors can be computed using numerical integration.

Summarizing the results in a table

	Posterior mean	Posterior SD	95% credible set
μ	0.17	1.31	(-2.49, 2.83)
σ	2.57	1.37	(1.10, 6.54)

Frequentist analysis of a normal mean

- ▶ In frequentist statistics, the estimate of the mean is $\bar{Y}$.
- ▶ If σ is known, the 95% interval is

$$\bar{Y} \pm z_{0.975} \frac{\sigma}{\sqrt{n}},$$

where z is the quantile of a normal distribution.

- ▶ If σ is unknown, the 95% interval is

$$\bar{Y} \pm t_{0.975, n-1} \frac{s}{\sqrt{n}},$$

where t is the quantile of a Student's t -distribution.

Bayesian analysis of a normal mean

- ▶ The Bayesian estimate of μ is its marginal posterior mean.
- ▶ The interval estimate is the 95% posterior credible interval.
- ▶ If σ is known, the posterior of $\mu|\mathbf{Y}$ is Gaussian (**check**) and the 95% credible interval is

$$E(\mu|\mathbf{Y}) \pm z_{0.975}SD(\mu|\mathbf{Y}).$$

- ▶ If σ is unknown, the marginal (over σ^2) posterior of μ is Student's t with $\nu = n + 2a$ degrees of freedom (**check**).
- ▶ Therefore, the 95% credible interval is

$$E(\mu|\mathbf{Y}) \pm t_{0.975,\nu}SD(\mu|\mathbf{Y}).$$

Case 1: σ is known

The posterior is

$$\begin{aligned} p(\mu|\mathbf{Y}) &\propto \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mu)^2 \right] \times \exp \left[-\frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 \right] \\ &\propto \exp \left[-\frac{n\mu^2}{2\sigma^2} \right] \times \exp \left[\frac{\mu \sum_{i=1}^n Y_i}{\sigma^2} \right] \times \exp \left[-\frac{\mu^2}{2\sigma_0^2} \right] \times \exp \left[\frac{\mu\mu_0}{\sigma_0^2} \right] \\ &= \exp \left[-\frac{1}{2}\mu^2 \left(\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) \right] \times \exp \left[\mu \left(\frac{\sum_{i=1}^n Y_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \right) \right]. \end{aligned}$$

By matching the terms with those in the density of a normal distribution, we have

$$\mu|\mathbf{Y} \sim \text{Normal} \left(\left[\frac{\sum_{i=1}^n Y_i}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \right] / \left[\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right], 1 / \left[\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right] \right).$$

Case 2: σ is unknown

The priors are: $\mu|\sigma^2 \sim \text{Normal}(\mu_0, c\sigma^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.

The posterior is

$$\begin{aligned} p(\mu, \sigma^2 | \mathbf{Y}) &\propto (\sigma^2)^{-n/2} \exp \left[-\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2\sigma^2} \right] \exp \left[-\frac{(\mu - \mu_0)^2}{2c\sigma^2} \right] (\sigma^2)^{-a-1} \exp \left[-\frac{b}{\sigma^2} \right] \\ &= (\sigma^2)^{-(a+n/2)-1} \exp \left[-\frac{1}{\sigma^2} \left\{ \frac{\sum_{i=1}^n (Y_i - \mu)^2}{2} + \frac{(\mu - \mu_0)^2}{2c} + b \right\} \right]. \end{aligned}$$

The density has the same kernel as an Inverse-Gamma distribution with shape parameter $a + n/2$ and rate parameter $\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2} + \frac{(\mu - \mu_0)^2}{2c} + b$. Thus, after integrating out with respect to σ^2 , we have

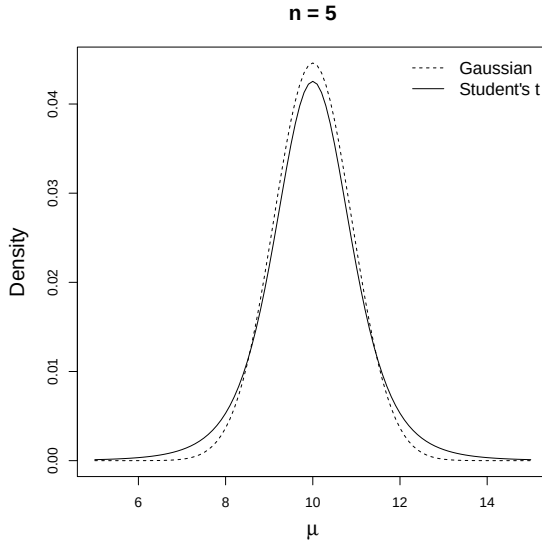
$$p(\mu | \mathbf{Y}) = \int p(\mu, \sigma^2 | \mathbf{Y}) d\sigma^2 \propto \left\{ \frac{\sum_{i=1}^n (Y_i - \mu)^2}{2} + \frac{(\mu - \mu_0)^2}{2c} + b \right\}^{-\frac{n+2a}{2}},$$

which is similar to the kernel of a Student's t distribution with df $n + 2a$.

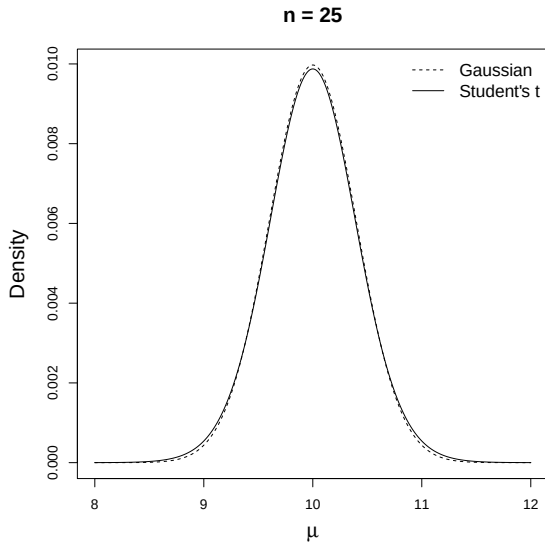
Bayesian analysis of a normal mean

- ▶ The following two slides give the posterior of μ for a data set with sample mean 10 and sample variance 4.
- ▶ The Gaussian analysis assumes $\sigma^2 = 4$ is known.
- ▶ The Student's t analysis integrates over uncertainty in σ^2 .
- ▶ As expected, the latter interval is a bit wider.

Bayesian analysis of a normal mean



Bayesian analysis of a normal mean



Example with two parameters

- ▶ Assume $N = 10$ patients are given treatment and $N = 10$ are given control.
- ▶ Let θ_1 and θ_2 be the survival probabilities for the control and treatment groups, respectively.
- ▶ We pick priors $\theta_1, \theta_2 \stackrel{iid}{\sim} \text{Uniform}(0, 1)$.
- ▶ We observe $Y_1 = 5$ survivals and $Y_2 = 8$ survivals for control and treatment groups, respectively.
- ▶ Our goal is to determine if treatment improves survival, i.e., $\theta_2 > \theta_1$.
- ▶ This requires summarizing the bivariate posterior $p(\theta_1, \theta_2 | Y_1, Y_2)$.

Methods for dealing with multiple parameters

- ▶ We want to compute $\text{Prob}(\theta_2 > \theta_1 | Y_1, Y_2)$.
- ▶ We could do an integral or grid approximation.
- ▶ Both often fail when there are many parameters.
- ▶ We need new tools!
- ▶ Monte Carlo sampling will be a key tool.
- ▶ We'll spend a month on this in the computing section.

Comparing proportions

- ▶ The model is
 - ▶ $Y_1|\theta_1 \sim \text{Binomial}(N, \theta_1)$
 - ▶ $Y_2|\theta_2 \sim \text{Binomial}(N, \theta_2)$
 - ▶ $\theta_1, \theta_2 \stackrel{\text{iid}}{\sim} \text{Beta}(1, 1)$
- ▶ The full posterior is

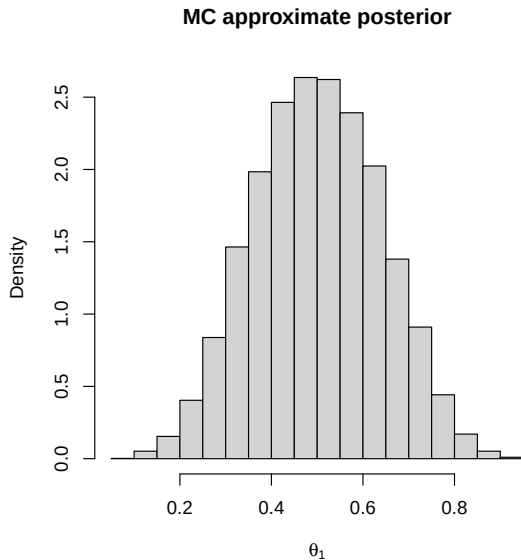
$$\begin{aligned} p(\theta_1, \theta_2 | Y_1, Y_2) &\propto f(Y_1, Y_2 | \theta_1, \theta_2) \pi(\theta_1, \theta_2) \\ &= f(Y_1 | \theta_1) f(Y_2 | \theta_2) \pi(\theta_1) \pi(\theta_2) \\ &= \{f(Y_1 | \theta_1) \pi(\theta_1)\} \{f(Y_2 | \theta_2) \pi(\theta_2)\}. \\ &\propto p(\theta_1 | Y_1) p(\theta_2 | Y_2). \end{aligned}$$

- ▶ The posterior is $\theta_1 | Y_1, Y_2 \sim \text{Beta}(Y_1 + 1, N - Y_1 + 1)$ independent of $\theta_2 | Y_1, Y_2 \sim \text{Beta}(Y_2 + 1, N - Y_2 + 1)$.
- ▶ The next slide approximates $\text{Prob}(\theta_2 > \theta_1 | Y_1, Y_2)$ using MC sampling.

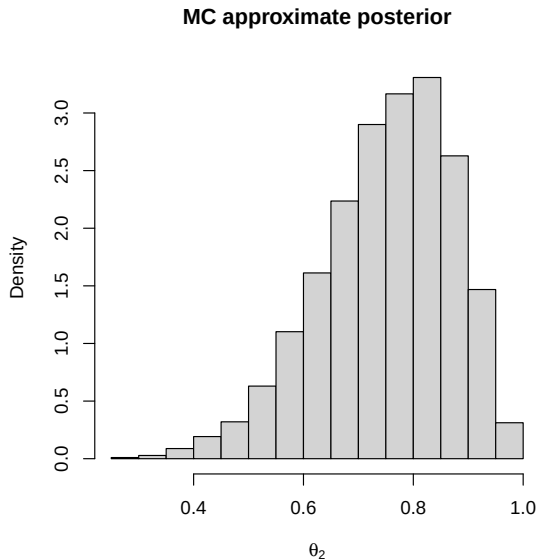
Summarizing a posterior using MC sampling

```
> N <- 10; Y1 <- 5; Y2 <- 8 # Data
> set.seed(100) # Keeps the results reproducible
> S <- 10000 # Number of MC samples
> theta1 <- rbeta(S,Y1+1,N-Y1+1); theta2 <- rbeta(S,Y2+1,N-Y2+1)
>
> hist(theta1,xlab=expression(theta[1]),
+       main="MC approximate posterior", probability = TRUE)
> (Y1+1)/(N+2) # True post mean
[1] 0.5
> mean(theta1) # MC estimate
[1] 0.5010727
> plot(theta1,theta2,xlim=c(0,1),ylim=c(0,1),
+       xlab=expression(theta[1]), ylab=expression(theta[2]))
> abline(a=0, b = 1) # adds diagonal line
> mean(theta2>theta1)
[1] 0.9103
```

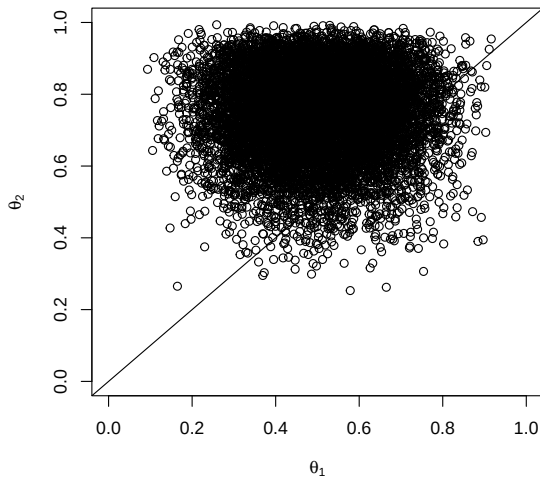
Marginal posterior, $p(\theta_1 | Y_1, Y_2)$



Marginal posterior, $p(\theta_2 | Y_1, Y_2)$



Joint posterior, $p(\theta_1, \theta_2 | Y_1, Y_2)$

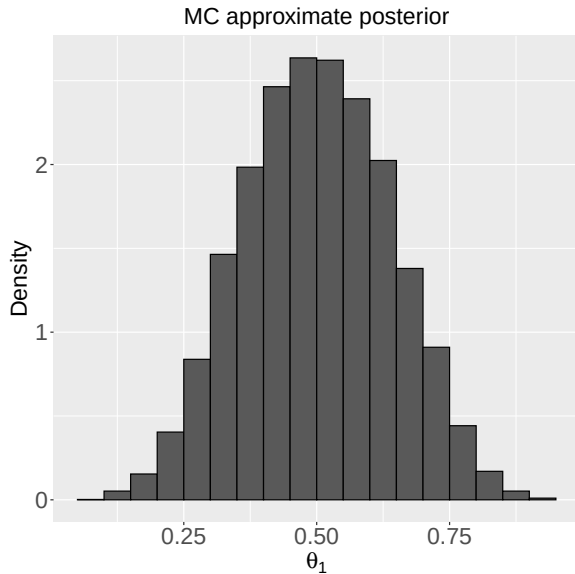


Supplementary slides for `ggplot`

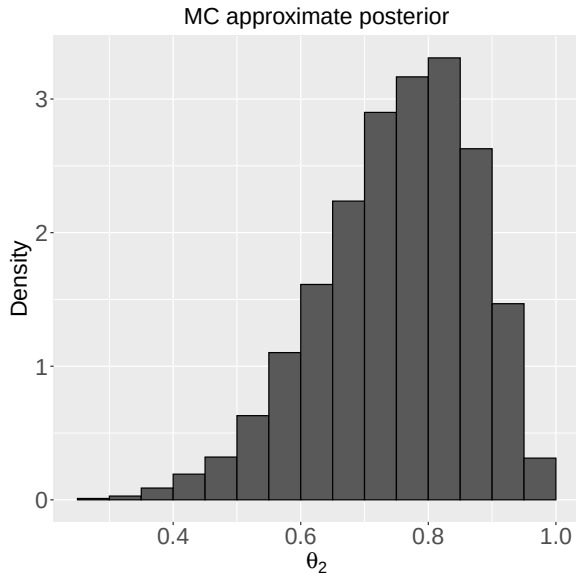
ggplot codes

```
p <- ggplot() +  
  geom_histogram(aes(x= theta1, y=..density..),  
                 breaks = hist(theta1, plot = F)$breaks,  
                 colour="black") +  
  ggtitle("MC approximate posterior") +  
  theme(plot.title = element_text(hjust = 0.5)) +  
  xlab(expression(theta[1])) + ylab("Density") +  
  theme(axis.text=element_text(size=20),  
        axis.title=element_text(size=20),  
        plot.title = element_text(size=20))  
ggsave("ggplot_pos_dist_theta1.pdf", p, height = 7, width = 7)
```

Marginal posterior, $p(\theta_1 | Y_1, Y_2)$



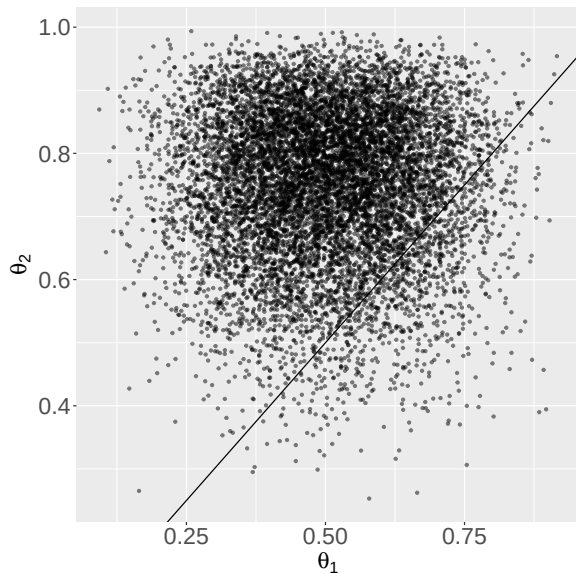
Marginal posterior, $p(\theta_2 | Y_1, Y_2)$



ggplot codes

```
p <- ggplot() + geom_point(aes(x = theta1, y = theta2),  
                             size = 1, alpha = 0.5) +  
  xlab(expression(theta[1])) + ylab(expression(theta[2])) +  
  theme(axis.text=element_text(size=20),  
        axis.title=element_text(size=20)) +  
  geom_abline(slope = 1)  
ggsave(p, filename = "ggplot_pos_dist_theta1_theta2.pdf",  
       width = 7, height = 7)
```

Joint posterior, $p(\theta_1, \theta_2 | Y_1, Y_2)$



Thank you!